

Experiment 4 – Implement svm and K-Nearest Neighbors (KNN) to Classify the Iris Dataset

Problem Statement 1

A botanist wants to develop a Machine Learning model to automatically classify Iris flowers into one of three species based on their flower measurements.

Use the standard **Iris dataset** available in Scikit-learn and implement the K-Nearest Neighbors (KNN) classification algorithm.

Task 1: Load the Dataset

Task 2: Data Preprocessing (Apply **StandardScaler**)

Task 3: Train the KNN Classifier

1. Create a KNN classifier with **k = 5**.
2. Train the classifier using the training data.
3. Predict the species of the test data.

Task 4: Analyze Predictions

Create a table showing:

Actual Class	Predicted Class	Prediction Status
Setosa	Setosa	Correct
Versicolor	Virginica	Wrong
Virginica	Virginica	Correct

Classify every prediction as either:

- **Correct**
- **Wrong**

Display:

- Total Correct Predictions
- Total Wrong Predictions

Task 5: Model Evaluation

Calculate:

- Accuracy
- Confusion Matrix
- Classification Report

Task 6: Visualization

Create a bar chart showing:

- Number of Correct Predictions
- Number of Wrong Predictions

- a) Why is feature scaling necessary before applying KNN?
- b) What is the role of the value of **k** in the KNN algorithm?
- c) What is overfitting in KNN? What happens if **k = 1**?
- d) Which Iris species is most frequently misclassified? Explain why.
- e) Why is the confusion matrix useful even when the model accuracy is high?

Problem Statement 2

A flower research laboratory wants to automatically classify Iris flowers into three different species using Machine Learning. The researchers would like to study how different values of the **regularization parameter (C)** and the **kernel coefficient (gamma)** affect the performance of the Support Vector Machine (SVM) classifier.

Use the standard **Iris dataset** available in Scikit-learn to build multiple SVM models, compare their performance, and identify the best parameter combination.

Task 1: Dataset Investigation

Task 2: Data Preprocessing

Task 3: Train SVM Models

Train SVM classifiers using the **RBF kernel** for the following combinations:

Model	C	Gamma
Model 1	0.1	0.1

Model 2	1	0.1
Model 3	10	0.1
Model 4	1	1
Model 5	1	10

For each model:

- Train the classifier.
- Predict the test dataset.
- Calculate the accuracy.

Task 4: Compare Predictions

For the **best-performing model**, create a table showing:

Sample	Actual Class	Predicted Class	Status
1	Setosa	Setosa	Correct
2	Virginica	Versicolor	Wrong
3	Versicolor	Versicolor	Correct

Display:

- Total Correct Predictions
- Total Wrong Predictions

Task 5: Model Evaluation

For the best model:

- Accuracy
- Confusion Matrix
- Classification Report

Task 6: Parameter Comparison

Prepare a comparison table.